

deep learning 2, 3

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1 PCA

Claim: Cyclicity of trace. Let A, B be $n \times m, m \times n$. Then $\text{tr}(AB) = \text{tr}(BA)$.

Proof: We have

$$\begin{aligned}\text{tr}(AB) &= \sum_{i \in [n]} (AB)_{ii} = \sum_{i \in [n]} \sum_{j \in [m]} A_{ij} B_{ji} \\ &= \sum_{j \in [m]} \sum_{i \in [n]} B_{ji} A_{ij} = \sum_{j \in [m]} (BA)_{jj} \\ &= \text{tr}(BA).\end{aligned}$$

Claim: General cyclicity of trace. Let $\{A_i\}_{i \in [k]}$ be $n_i \times n_{i+1}$, where $n_{k+1} = n_1$. Then $\text{tr}\left(\prod_{i=1}^k A_i\right) = \text{tr}\left(A_k \prod_{i=1}^{k-1} A_i\right)$.

Proof: Let $B = \prod_{i=1}^{k-1} A_i$. The result becomes $\text{tr}(BA_k) = \text{tr}(A_k B)$.

Optimal matrix for PCA. Let $\mathbb{R}^n$ be example space, $\mathbb{R}^l$ be code space, $D : \mathbb{R}^l \rightarrow \mathbb{R}^n$ be decoder, so $D \in \mathbb{R}^{n \times l}$. Let $\{\mathbf{x}^{(i)}\}_{i \in [m]}$ denote the samples. We wish to compute

$$\begin{aligned}D^* &= \text{argmin}_D \sum_{ij} (x_j^i - (r(x^i))_j)^2 \\ &= \text{argmin}_D \sum_i \sum_j (x_j^i - (r(x^i))_j)^2 = \sum_i \|x^i - r(x^i)\|_2^2.\end{aligned}$$

Suppose $l = 1$, so $D = d$. Then

$$\begin{aligned}
d^* &= \operatorname{argmin}_d \sum_i \|x^i - dd^T x^i\|_2^2 = \operatorname{argmin}_d \|X^T - dd^T X^T\|_2^2 \\
&= \operatorname{argmin}_d \|X - Xdd^T\|_2^2 = \operatorname{argmin}_d \operatorname{tr}(X^T - dd^T X^T)(X - Xdd^T) \\
&= \operatorname{argmin}_d \operatorname{tr}(X^T X - dd^T X^T X - X^T X dd^T + dd^T X^T X dd^T) \\
&= \operatorname{argmin}_d -2\operatorname{tr}(X^T X dd^T) + \operatorname{tr}(X^T X dd^T dd^T) \\
&= \operatorname{argmin}_d -2\operatorname{tr}(X^T X dd^T) + \operatorname{tr}(X^T X dd^T) \\
&= \operatorname{argmin}_d -\operatorname{tr}(X^T X dd^T) \\
&= \operatorname{argmax}_d \operatorname{tr}(d^T X^T X d)
\end{aligned}$$

Thus, we take d to be an eigenvector corresponding to the largest eigenvalue of $X^T X$ (which is symmetric and therefore diagonalizable).

More generally,

$$\begin{aligned}
D^* &= \operatorname{argmin}_D \sum_{ij} (x_j^i - r(x^i)_j)^2 = \operatorname{argmin}_D \sum_i \|x^i - r(x^i)\|^2 \\
&= \operatorname{argmin}_D \sum_i \|x^i - DD^T x^i\|^2 = \operatorname{argmin}_D \|X - XDD^T\|_2^2
\end{aligned}$$

Claim: Let $V = n \times m$ and $A = n \times n$. Then $\operatorname{tr}(V^T AV) = \sum_{i \in [m]} v_i^T A v_i$.

Proof: We have

$$\operatorname{tr}(V^T AV) = \sum_{i \in [m]} (V^T AV)_{ii} = \sum_{i \in [m]} v_i^T A v_i,$$

as desired.

2 Probability and Information Theory

Let X be a continuous RV with $I_2 < \infty$, mean μ , and variance σ^2 . Note

$$\mu^2 = \left(\int_{-\infty}^{\infty} xp(x)dx \right)^2 \leq \int_{-\infty}^{\infty} x^2 p(x)dx \int_{-\infty}^{\infty} p(x)dx = I_2,$$

so $\mu < \infty$. Then

$$\begin{aligned}
\sigma^2 &= \int_{-\infty}^{\infty} (x - \mu)^2 p(x)dx = \int_{-\infty}^{\infty} x^2 p(x)dx - 2\mu \int_{-\infty}^{\infty} xp(x)dx + \mu^2 \\
&= I_2 - \mu^2
\end{aligned}$$

so $\sigma < \infty$ as well.

I claim that $X = \mathcal{N}(x, \mu, \sigma^2)$ maximizes entropy for given μ, σ^2 .

Entropy.

$$\begin{aligned}
H(X|Y) &= \mathbb{E}_{y \sim Y}[H(X|Y = y)] = \sum_y H(X|Y = y)p_Y(y) \\
&= - \sum_y p_Y(y) \sum_x p_{X|Y}(x, y) \log p_{X|Y}(x, y) \\
&= - \sum_y p_Y(y) \sum_x \frac{p_{XY}(x, y)}{p_Y(y)} \log \frac{p_{XY}(x, y)}{p_Y(y)} \\
&= - \sum_y p_Y(y) \sum_x \frac{p_{XY}(x, y)}{p_Y(y)} (\log p_{XY}(x, y) - \log p_Y(y)) \\
&= - \sum_y \sum_x p_{XY}(x, y) (\log p_{XY}(x, y) - \log p_Y(y)) \\
&= H(X, Y) - H(Y).
\end{aligned}$$

Let f be a function. Then $H(X) \geq H(f(X))$. Let $S = \text{im } f$. Then

$$\begin{aligned}
H(X) &= \sum_{y \in S} \sum_{x \in f^{-1}y} -p_X(x) \log p_X(x) \\
&\leq \sum_{y \in S} -p_Y(y) \log p_Y(y) = H(f(X)).
\end{aligned}$$

Multinomial distribution. We have $p(x) = \frac{1}{\sqrt{(2\pi)^n \det \Sigma}} \exp((x - \mu)^T \Sigma^{-1} (x - \mu))$.

Write $\Sigma = QD^2Q^T$, where D is invertible and Q is orthogonal. Then

$$\begin{aligned}
\int_{\mathbb{R}^n} p(x) dX &= \int_{\mathbb{R}^n} \frac{1}{\sqrt{(2\pi)^n \det \Sigma}} \exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1} (x - \mu)\right) dX \\
&= \int_{\mathbb{R}^n} \frac{1}{\sqrt{(2\pi)^n \det \Sigma}} \exp\left(-\frac{1}{2}(x - \mu)^T QD^{-2}Q^T (x - \mu)\right) dX.
\end{aligned}$$

Let $y = D^{-1}Q^T(x - \mu)$, so $dY = \det(D^{-1}Q^T)dX$, so $dX = \det(QD)dY$. Thus,

$$\begin{aligned}
\int_{\mathbb{R}^n} p(x)dX &= \int_{\mathbb{R}^n} \frac{1}{\sqrt{(2\pi)^n \det \Sigma}} \exp\left(-\frac{1}{2}y^T y\right) \det(QD)dY \\
&= \frac{1}{(2\pi)^{n/2}} A_{n-1} \int_{r=0}^{\infty} r^{n-1} e^{-r^2/2} dr \\
&= \frac{1}{(2\pi)^{n/2}} A_{n-1} \int_{u=0}^{\infty} (2u)^{(n-1)/2} e^{-u} (2u)^{-1/2} du \\
&= \frac{1}{(2\pi)^{n/2}} A_{n-1} \int_{u=0}^{\infty} (2u)^{n/2-1} e^{-u} du \\
&= \frac{2^{n/2-1}}{(2\pi)^{n/2}} A_{n-1} \Gamma(n/2) = \frac{A_{n-1} \Gamma(n/2)}{2\pi^{n/2}} = 1.
\end{aligned}$$

$\log p(X, Y|Z) = \log p(X|Z) + \log p(Y|X, Z)$. We have

$$\begin{aligned}
p(X, Y|Z) &= \frac{p(X, Y, Z)}{p(Z)} = \frac{p(X, Z)}{p(Z)} \cdot \frac{p(X, Y, Z)}{p(X, Z)} \\
&= p(X|Z) \cdot p(Y|X, Z)
\end{aligned}$$

Let $(X, Y) \sim p(x, y) = p(x)p(y|x)$. Then $I(X, Y)$ is concave in $p(x)$ for fixed $p(y|x)$ and convex in $p(y|x)$ for fixed $p(x)$.

Proof: Suppose $p(y|x)$ is fixed. Write $I(X, Y) = H(Y) - H(Y|X)$. $H(Y)$ is concave in $p(y)$. $p(y) = \sum_x p(y|x)p(x)$ linear in $p(x)$, so $H(Y)$ is concave in $p(x)$. $H(Y|X) = \sum_x H(Y|X=x)p(x)$ and $H(Y|X=x) = -\sum_y p(y|x) \log p(y|x)$ is constant, so $H(Y|X)$ is linear in $p(x)$. Thus, $I(X, Y)$ is concave in $p(x)$.

Expanding, we have

$$\begin{aligned}
I(X, Y) &= H(Y) - H(Y|X) = -\sum_y p(y) \log p(y) - \sum_x H(Y|X=x)p(x) \\
&= -\sum_y \left(\sum_x p(y|x)p(x) \right) \log \left(\sum_x p(y|x)p(x) \right) + \sum_x \sum_y [p(y|x) \log p(y|x)]p(x),
\end{aligned}$$

so this as a function of $p(y|x), p(x)$ is concave in $p(x)$ for fixed $p(y|x)$.

3 Data Processing Inequality

Suppose we have $X \rightarrow Y \rightarrow Z$, so $p(x, y, z) = p(x)p(y|x)p(z|y)$. I claim that the conditional distribution of Z only depends on Y , so $p(z|y) = p(z|x, y)$.

Proof: We have $p(x, y, z)/p(x, y) = p(z|x, y) = p(z|y)$.

Claim: $X \rightarrow Y \rightarrow Z$ implies $Z \rightarrow Y \rightarrow X$.

Proof: Markovity is equivalent to $p(x, z|y) = p(x|y)p(z|y)$, which is symmetric in x, z .

Claim: If $Z = f(Y)$, then $X \rightarrow Y \rightarrow Z$.

Proof: Suppose $z \neq f(y)$. Then $p(x, z|y) = 0$ and $p(z|y) = 0$, so $p(x, z|y) = p(x|y)p(z|y)$. Suppose $z = f(y)$. Then $p(x, z|y) = p(x|y)$ and $p(z|y) = 1$, so $p(x, z|y) = p(x|y)p(z|y)$.

Claim: Data processing inequality. No processing of Y can increase the amount of information Y has about X . That is, if $X \rightarrow Y \rightarrow Z$ is Markov, then $I(X, Y) \geq I(X, Z)$.

Proof: First, a lemma. Let X, Y, Z be arbitrary rvs. Then $H(X|Y) \geq H(X|Y, Z)$. Indeed,

$$\begin{aligned} -H(X|Y, Z) &= \sum_{x,y,z} p(x, y, z) \log p(x|y, z) = \sum_{x,y,z} p(x, y, z) \log \frac{p(x, y, z)}{p(y, z)} \\ &= \sum_{x,y} \sum_z p(x, y, z) \log \frac{p(x, y, z)}{p(y, z)} \geq \sum_{x,y} p(x, y) \log \frac{p(x, y)}{p(y)} \\ &= \sum_{x,y} p(x, y) \log p(x|y) = -H(X|Y), \end{aligned}$$

as desired. We wish to show $H(X) + H(Y) - H(X, Y) \geq H(X) + H(Z) - H(X, Z)$, or equivalently, $H(Y) + H(X, Z) \geq H(Z) + H(X, Y)$. Since $X \rightarrow Y \rightarrow Z$ is Markov, we have $p(y)p(x, y, z) = p(x, y)p(y, z)$. Taking log-expectancy, we have $H(Y) + H(X, Y, Z) = H(X, Y) + H(Y, Z)$. Adding $H(Y, Z)$, we have $H(X, Z) + H(Y, Z) \geq H(Z) + H(X, Y, Z)$, or $H(X, Z) - H(Z) \geq H(X, Y, Z) - H(Y, Z)$, or $H(X|Z) \geq H(X|Y, Z)$, which follows from the lemma above.

Book proof: Consider $I(X; Y, Z) = I(X; Y) + I(X; Z|Y) = I(X; Z) + I(X; Y|Z)$. Since $I(X; Z|Y) = 0$ (since X, Z are conditionally independent given Y), we have $I(X; Y) \geq I(X; Z)$.

Claim: $I(X; Y) \geq I(X; g(Y))$.

Proof: Consider the Markov chain $X \rightarrow Y \rightarrow g(Y)$. We have $I(X; Y) \geq I(X; g(Y))$.

Claim: If $X \rightarrow Y \rightarrow Z$, then $I(X; Y|Z) \leq I(X; Y)$.

Proof: Obvious from above result.

Claim: Let $p, q \in I$. Then $D(\text{Bern}(p) \parallel \text{Bern}(q)) \geq 2(p-q)^2$ (log taken in base e).

Proof: $D(\text{Bern}(p) \parallel \text{Bern}(q))$ is given by $f(p, q) = p \log \frac{p}{q} + (1-p) \log \frac{1-p}{1-q}$. We compute

$$\begin{aligned} \frac{\partial f}{\partial p} &= \log \frac{p}{q} + p \cdot \frac{q}{p} \cdot \frac{1}{q} + \log \frac{1-p}{1-q} + (1-p) \cdot \frac{1-q}{1-p} \cdot \frac{-1}{1-q} \\ &= \log \frac{p(1-q)}{q(1-p)} = \log p - \log(1-p) + \log \frac{1-q}{q} \\ \frac{\partial^2 f}{\partial p^2} &= \frac{1}{p} - \frac{1}{1-p}(-1) = \frac{1}{p} + \frac{1}{1-p} \geq 2, \end{aligned}$$

where the inequality follows from AM-HM. Note that $\frac{\partial f}{\partial p}(q, q) = \log 1 = 0$. Suppose $p \neq q$. By Taylor Series with Remainder, there exists p' between p, q such that

$$\begin{aligned} f(p, q) &= f(q, q) + \frac{\partial f}{\partial p}(q, q)(p-q) + \frac{\partial^2 f}{\partial p^2}(p', q)(p-q)^2 \\ &\geq 2(p-q)^2, \end{aligned}$$

as desired.

Let P, Q be distributions on X . Define $\text{TV}(P, Q) \triangleq \sup_{E \subseteq X} P(E) - Q(E)$. Then Pinsker-Csiszar states

$$\text{TV}(P, Q) \leq \sqrt{\frac{1}{2} D(P \parallel Q)},$$

(where log is taken base E).

Proof: Fix $E \subseteq X$ and let $p = P(E), q = Q(E)$. I claim that $D(P \parallel Q) \geq D(\text{Bern}(p) \parallel \text{Bern}(q))$. Indeed, by the log-sum inequality, we have

$$\begin{aligned} D(P \parallel Q) &= \sum_x p(x) \log \frac{p(x)}{q(x)} = \sum_{x \in E} p(x) \log \frac{p(x)}{q(x)} + \sum_{x \notin E} p(x) \log \frac{p(x)}{q(x)} \\ &\geq p(E) \log \frac{p(E)}{q(E)} + (1-p(E)) \log \frac{1-p(E)}{1-q(E)} \\ &= D(\text{Bern}(p) \parallel \text{Bern}(q)), \end{aligned}$$

as desired. It follows that $D(P \parallel Q) \geq 2(p-q)^2$, so $\sqrt{\frac{1}{2} D(P \parallel Q)} \geq 2(P(E) - Q(E))$ for all E , so $\sqrt{\frac{1}{2} D(P \parallel Q)} \geq \text{TV}(P, Q)$.

4 Numerical Optimization

Claim: Let $A \in \mathbb{R}^{n \times n}$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ by $f(x) = x^T A x$. Then $\nabla f = A^T x + A x$.

Proof: Write

$$\begin{aligned} f(x) &= x^T A x = x^T \left(\sum_j A_{ij} x_j \right) \\ &= \sum_i x_i \sum_j A_{ij} x_j = \sum_{ij} A_{ij} x_i x_j, \end{aligned}$$

so

$$\begin{aligned} \frac{\partial f}{\partial x_k} &= \sum_i A_{ik} x_i + \sum_j A_{kj} x_j \\ &= A[:, k]^T x + A[k, :] x \\ \nabla f &= A^T x + A x \end{aligned}$$

In particular, if A is symmetric, then $\nabla f = 2Ax$.

Claim: Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ has continuous second order partials with Lipschitz Hessian $H(f)$ at x_0 . Then f can be approximated by a second order Taylor series.

Proof: Let $E(x) = f(x) - (f(x_0) + (x - x_0)^T \nabla f(x_0) + \frac{1}{2}(x - x_0)^T H(f)(x_0)(x - x_0))$. Then

$$\begin{aligned} \nabla E(x) &= \nabla f(x) - \nabla f(x_0) - H(f)(x_0)(x - x_0) \\ J(\nabla E)(x) &= H(f)(x) - H(f)(x_0) \end{aligned}$$

Note that we invoked that $H(f)(x_0)$ is symmetric since the second order partials are continuous. Since $H(f)$ is Lipschitz at x_0 , let $|H(f)(x) - H(f)(x_0)| \leq L$ for $|x - x_0| \leq \delta$. Then

$$\nabla E(x) = \int_{\gamma} J(\nabla E)(x) dx \rightarrow |\nabla E(x)| \leq \int_{\gamma} L|x - x_0| |dx|$$

Taking γ to be the straight-line path from x_0 to x , we have $|\nabla E(x)| \leq \frac{1}{2}L|x - x_0|^2$. Similarly,

$$E(x) = \int_{\gamma} \nabla E(x) dx \rightarrow |E(x)| \leq \int_{\gamma} \frac{1}{2}L|x - x_0|^2 |dx|,$$

so $|E(x)| \leq \frac{1}{6}L|x - x_0|^3$, as desired.

Claim: Let f have second order partials with Lipschitz Hessian $H(f)$ at x_0 ,

$\nabla f(x_0) = 0$, and $H(f)(x_0)$ is positive definite. Then x_0 is a local minimum of f .

Proof: Let $\lambda_{\min} > 0$ denote the smallest eigenvalue of $H(f)(x_0)$. Pick $|x - x_0| \leq \frac{\lambda_{\min}}{L}$. Then

$$\begin{aligned} f(x) &\geq f(x_0) + \frac{1}{2}(x - x_0)^T H(f)(x_0)(x - x_0) - \frac{1}{6}L|x - x_0|^3 \\ &\geq f(x_0) + \frac{1}{2}|x - x_0|^2 \lambda_{\min} - \frac{1}{6}L|x - x_0|^3 \\ &= f(x_0) + \frac{1}{6}|x - x_0|^2 (3\lambda_{\min} - L|x - x_0|) \\ &\geq f(x_0) + \frac{1}{3}|x - x_0|^2 \lambda_{\min}, \end{aligned}$$

so x_0 is indeed a local minimum.

Claim: Suppose that $f : \mathbb{R}^n \rightarrow \mathbb{R}$ has continuous second partials and has a local minimum at x_0 . Then $H(f)(x_0)$ is positive definite.

Proof: Let $u \in \mathbb{R}^n$ be a unit vector. Define $g : \mathbb{R} \rightarrow \mathbb{R}$ by $g(t) = f(x_0 + tu)$, so g has a local minimum at $t = 0$. Then $g'(t) = u^T \nabla f(x_0 + tu)$ and $g''(t) = u^T ((J \nabla f)(x_0 + tu)u) = u^T H(f)(x_0 + tu)u$, so $g''(0) = u^T H(f)(x_0)u > 0$. Since this holds for all u , $H(f)(x_0)$ is positive definite.

Claim: (Newton's Method). Let $f(x) = f(x_0) + (x - x_0)^T \nabla f(x_0) + \frac{1}{2}(x - x_0)^T H(x - x_0)$. Then f is minimal when $x^* = x_0 - H^{-1} \nabla f(x_0)$ (for positive semidefinite H).

Proof: Write $H = Q^T \Lambda Q$ for Q orthogonal. Then

$$\begin{aligned} x^* &= \operatorname{argmin}_x (x - x_0)^T \nabla f(x_0) + \frac{1}{2}(x - x_0)^T Q^T \Lambda Q (x - x_0) \\ &= \operatorname{argmin}_x (Q(x - x_0))^T Q \nabla f(x_0) + \frac{1}{2}(Q(x - x_0))^T \Lambda Q (x - x_0) \\ &= x_0 + Q^T \operatorname{argmin}_y y^T Q \nabla f(x_0) + \frac{1}{2}y^T \Lambda y \\ &= x_0 + Q^T \operatorname{argmin}_y \sum_{i=1}^n (Q \nabla f(x_0))_i y_i + \frac{1}{2} \lambda_i y_i^2 \\ &= x_0 + Q^T \Lambda^{-1} (-Q \nabla f(x_0)) \\ &= x_0 - H^{-1} \nabla f(x_0) \end{aligned}$$

The argmin was removed by minimizing the objective component-wise.

5 Norms

The l_p norm on $\mathbb{R}^{n \times m}$ is defined to be $\|X\|_p = \sup_{y, y \neq 0} \frac{\|Xy\|_p}{\|y\|_p}$. Note that l_2 doesn't equal the Frobenius norm, since $\|I\|_F = \sqrt{n}$, while $\|X\|_2 = 1$.

Claim: l_p is indeed a norm.

Proof: (1). Suppose $\|X\|_p = 0$. Then $\|Xy\|_p = 0$ for all $y \neq 0$, so $Xy = 0$ for all y , so $Xe_i = 0$ for all i , so $X = 0$. (2). We show $\|X + Y\|_p \leq \|X\|_p + \|Y\|_p$. We have

$$\begin{aligned} \|X + Y\|_p &= \sup_{y, y \neq 0} \|(X + Y)y\|_p / \|y\|_p \\ &= \sup_{y, y \neq 0} \|Xy + Yy\|_p / \|y\|_p \\ &\leq \sup_{y, y \neq 0} \|Xy\|_p / \|y\|_p + \sup_{y, y \neq 0} \|Yy\|_p / \|y\|_p \\ &= \|X\|_p + \|Y\|_p. \end{aligned}$$

Since $\|X\|_p = \|-X\|_p$, this norm is clearly symmetric. $\|X\|_p$ is well-defined since we can take $\|y\|_p = 1$, and that the set of such y form a compact subset of $\mathbb{R}^n$.

Claim: $\|X\|_2 = \sigma_{\max}(X) = \sqrt{\lambda_{\max}(X^T X)}$.

Proof: Write $X = U\Sigma V^T$, let X have rank r , $U = [u_1, u_2, \dots, u_m]$, $V = [v_1, v_2, \dots, v_n]$. Write $y = \sum_{i=1}^n a_i v_i$. Then

$$\begin{aligned} \|Xy\|_2 / \|y\|_2 &= \left\| \sum_{i=1}^r \sigma_i u_i v_i^T y \right\|_2 / \|y\|_2 \\ &= \left\| \sum_{i=1}^r a_i \sigma_i u_i \right\|_2 / \left\| \sum_{i=1}^n a_i v_i \right\|_2 \\ &= \sqrt{\sum_{i=1}^r (a_i \sigma_i)^2} / \sqrt{\sum_{i=1}^n a_i^2} \\ &\leq \sigma_{\max} \sqrt{\sum_{i=1}^n a_i^2} / \sqrt{\sum_{i=1}^n a_i^2} = \sigma_{\max} \end{aligned}$$

Equality is achievable.

Let $\|\cdot\|$ be a norm on $\mathbb{R}^n$. Its dual norm is $\|x\|_* = \sup\{x^T y : \|y\| \leq 1\}$.

Proof of correctness: Suppose $x \neq 0$. Pick $y = x/\|x\|$, so $x^T y = x^T x / \|x\| = \|x\| > 0$. The metric is symmetric. Finally, $\|x + y\|_* = \sup\{(x + y)^T z : \|z\| \leq 1\} \leq$

$$\sup\{x^T z : \|z\| \leq 1\} + \sup\{y^T z : \|z\| \leq 1\} = \|x\|_x + \|y\|_*.$$

Def: A cone $K \subseteq \mathbb{R}^n$ is a subset such that $0 \in K$, $\lambda x \in K$ for all $\lambda \geq 0$ and $x \in K$, and K is convex.

The dual cone K^* is given by $K^* = \{y \in \mathbb{R}^n : y^T x \geq 0 \forall x \in K\}$. I claim that K^* is indeed a cone. We have $0 \in K^*$. Suppose $y_1, y_2 \in K^*$. Then

$$y_3^T x = ((1 - \lambda)y_1 + \lambda y_2)^T x = (1 - \lambda)y_1^T x + \lambda y_2^T x \geq 0$$

Suppose $y \in K^*$ and $\lambda \geq 0$. Then $(\lambda y)^T x = \lambda y^T x \geq 0$, so $\lambda y \in K^*$.